

ABOUT ME

Biomedical student building intelligent systems combining embedded sensing, machine learning, and AI-driven decision pipelines, with projects spanning biomedical signal analysis, healthcare, and other AI applications. Interested in applied AI/ML, multimodal data fusion, agentic systems for real-world decision making.

EDUCATION

New Prince Shri Bhavani Senior Secondary School, Chennai	Class X - CBSE	2020-21	94.8%
	Class XII - CBSE	2022-23	96.8%
SSN College of Engineering, Chennai	B.E. Biomedical Engineering	2023-27	CGPA: 8.916 / 10
Indian Institute Of Technology, Madras	B.S Data science (Foundation)	2024-25	CGPA: 8.13 / 10

TECHNICAL SKILLS

- Programming:** Python, Java, MATLAB, JavaScript, C/C++
- Frameworks & Web:** React, FastAPI, REST APIs, Gradio, Streamlit
- AI/ML:** CNN, Transfer Learning, Computer Vision, LLM Integration, PyTorch, TensorFlow, scikit-learn, Librosa
- Data:** MongoDB, SQL, PostgreSQL, Statistical data analysis, Signal processing
- Hardware:** Arduino, ESP32, IMU Sensors, Embedded systems
- Tools & Platforms:** Git, GitHub, Hugging Face Spaces, VS Code

EXPERIENCE

- Biomedical Engineering Intern - Medway Group of Hospitals** (Clinical Exposure) **December 2025**
- Shadowed clinical biomedical engineers across Pulmonology, ICU, CICU, OT, Radiology, Cath Lab and Gynecology departments; observed maintenance and troubleshooting of medical equipment in live clinical settings.
- Research Intern - Indian Institute of Technology, Kanpur** (SURGE intern) **May 2026 - July 2026**
- Worked at Lab of Neural systems, under the guidance of professor, **Dr. Nitin Gupta**, in the Department of Biological Sciences and Bioengineering and contributed to a project investigating behavioural responses of Drosophila melanogaster.

PROJECTS

- KNEEGUARD - Patellar Stability Monitoring System** (Wearable IoT Device) **July-November 2025**
- Wearable Sensors · Edge Analytics · ESP32 · IoMT*
- Engineered an ESP32-based wearable knee brace integrating IMU and piezoelectric sensors for real-time patellar stability monitoring with edge analytics and live telemetry.
- FIBER-OPTIC BASED PHOTOTHERAPY DEVICE** **Ongoing**
- Biomedical Device Design · Wavelength Control · Prototype Integration*
- Designing a targeted UVB delivery system with wavelength control and dose calibration to minimize exposure to healthy tissue in dermatological treatment
- RESPISENSE - Multimodal Intelligent Respiratory Profiler** (K!Hacks 3.0 Hackathon) **December 2025**
- Signal Analytics · Multimodal Fusion · Deep Learning · IoMT*
- Developed an AI-powered multimodal respiratory monitoring system integrating physiological, acoustic, and environmental signals for accessible, real-time respiratory health assessment.
- SIGLEARN - Biomedical Signal Analysis & Learning Platform** **June 2026**
- Machine Learning · Biomedical Signal Processing · Hugging Face*
- Developed an AI-powered biomedical signal analysis platform featuring automated preprocessing, machine learning classification, interactive visualization, and a context-aware AI tutor for educational signal interpretation.

OTHER ACTIVITIES

- Vice Chair - SSN IEEE Photonics Society (2026)
- Publications Volunteer - ICBSII 2025, ICBSII 2026
- Newsletter Editor - SSN ABE (2024-present)
- Creative & PR Team - SSN LAKSHAYA (2024)
- Social Media Representative - IEEE STUDENT BRANCH (2024)
- Public speaker, Amateur writer and poet